

Syllabus

Calculus

Teacher Information

- Email: dt323259@gmail.com
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Books

- Advanced Calculus 2nd Ed. by Patrick M Fitzpatrick (Provided by Teacher)

Grading

- Homework 50% (Homework is due the week after it is assigned)
- Tests 30%
- Midterm/Final 20% (Also replaces your lowest test score for the semester)
- Extra Credit 5% (You may receive up to 5% extra credit on each test/homework)

Schedule

- Week 1
 - Preliminaries
 - Homework 1: Exercises for Appendix A (Page 564)
 - Extra Credit: Prove Proposition A.3 (Page 561)
- Week 2
 - Sections 1.1 (Completeness Axiom)
 - Homework 2: Section 1.1 1-9
 - Extra Credit: Section 1.1 17

- Week 3
 - Sections 1.2 & 1.3 (Distribution of Ints and Rationals & Inequalities and Identities)
 - Homework 3: Section 1.2 1-4, Section 1.3 1-6, 11, 21
 - Extra Credit: Section 1.3 22
- Week 4
 - Review Chapter 1
 - Chapter 1 Test
- Week 5
 - Section 2.1 (Convergence of Sequences)
 - Homework 4: Section 2.1 1-9
 - Extra Credit: Section 2.1 14
- Week 6
 - Sections 2.2 & 2.3 (Sequences and Sets & Monotone Convergence Thm)
 - Homework 5: Section 2.2 1-4, Section 2.3 1-4, 8
 - Extra Credit Section 2.2 5
- Week 7
 - Section 2.4 (Sequential Compactness Thm)
 - Homework 6: Section 2.4 1-9
 - Extra Credit: Section 2.4 12
- Week 8
 - Review Chapter 2
 - Chapter 2 Test
- Week 9
 - Section 3.1 (Continuity)
 - Homework 7: Section 3.1 1-9
 - Extra Credit: Section 3.1 14
- Week 10
 - Sections 3.2 & 3.3 (Extreme Value Thm & Intermediate Value Thm)
 - Homework 8: Section 3.2 1-4, Section 3.3 1-5
 - Extra Credit: Section 3.3 10
- Week 11
 - Sections 3.4 & 3.5 (Uniform Continuity & Epsilon Delta Continuity Criterion)
 - Homework 9: Section 3.4 1-5, Section 3.5 1-5
 - Extra Credit: Section 3.5 6

- Week 12
 - Sections 3.6 & 3.7 (Images, Inverses, and Monotone Functions & Limits)
 - Homework 10: Section 3.6 1-6, Section 3.7 1-5
 - Extra Credit: 3.7 9
- Week 13
 - Chapter 3 Review
 - Chapter 3 Test
- Week 14
 - Midterm
- Week 15
 - Section 4.1 (Algebra of Derivatives)
 - Homework 11: Section 4.1 1-10
 - Extra Credit: Section 4.1 19
- Week 16
 - Sections 4.2 & 4.3 (Differentiating Inverses and Compositions & Mean Value Thm)
 - Homework 12: Section 4.2 1-7, Section 4.3 1-7, 9
 - Extra Credit: 4.2 9
- Week 17
 - Sections 4.4 & 4.5 (Cauchy Mean Value Thm & Leibniz Notation)
 - Homework 14: Section 4.4 1-7, Section 4.5 1-3
 - Extra Credit: Section 4.4 8
- Week 18
 - Chapter 4 Review
 - Chapter 4 Test
- Week 19
 - Section 6.1 (Darboux Sums; Upper and Lower Integrals)
 - Homework 15: Section 6.1 1-6
 - Extra Credit: Section 6.1 7
- Week 20
 - Section 6.2 (Archimedes-Riemann Thm)
 - Homework 16: Section 6.2 1-7
 - Extra Credit: Section 6.2 11
- Week 21
 - Section 6.3 (Additivity, Monotonicity, Linearity)
 - Homework 17: Section 6.3 1,3,4,5
 - Extra Credit: Section 6.3 6

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- Week 22
 - Section 6.4 (Continuity and Integrability)
 - Homework 18: Section 6.4 1-6
 - Extra Credit: Section 6.4 9
 - Week 23
 - Section 6.5 (Integrating Derivatives)
 - Homework 19: Section 6.5 1-4
 - Extra Credit: Section 6.5 6
 - Week 24
 - Sections 6.6 (Differentiating Integrals)
 - Test 17: Section 6.6 1-3,5,6
 - Extra Credit: Section 6.6 12
 - Week 25
 - Chapter 6 Review
 - Chapter 6 Test
 - Week 26
 - Final